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Analysis of Return to Sport and Weight Training After Repair of the Pectoralis Major Tendon

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Background: The use of routine patient-reported outcomes after repair of the pectoralis major tendon (PMT) is often prone to the ceiling effect owing to the high functional demand of those who sustain this injury.

Hypothesis: A significant number of patients are expected to fail to achieve return to preoperative activity after PMT repair despite achieving significant improvements in functional score.

Study Design: Case series; Level of evidence, 4.

Methods: A prospectively maintained institutional database was reviewed for all patients undergoing PMT repair from 2010 to 2016. Patients were surveyed with regard to pre- and postoperative participation in sports, level of intensity, maximum weight repetitions in exercises utilizing the PMT, and American Shoulder and Elbow Surgeons (ASES) and Single Assessment Numeric Evaluation (SANE) scores.

Results: Forty-four patients (73.3%) were available for final follow-up. All patients were male. Mean ± SD follow-up was 51.1 ± 24.1 months. Mean age was 39.6 ± 8.8 years (range, 24-61 years), and mean body mass index was 28.6 ± 3.5 kg/m^2. The dominant side was affected in 20 of 44 cases. Acute repairs were performed in 30 cases and chronic in 14. There were statistically significant improvements in both ASES and SANE scores ($P < .001$). Return to sport at any level was achieved by 43 of 44 (97.7%) patients, while 22 of 44 patients (50.0%) reported returning to sport at the same or better intensity as preinjury status. On average, there was a 23.3% ± 45.6% decrease in 1–repetition maximum (1RM) barbell bench press, 14.7% ± 62.3% decrease in 5–repetition maximum barbell bench press, 24.3% ± 21.8% decrease in 1RM dumbbell bench press, 35.7% ± 32.1% decrease in 1RM dumbbell fly, and 15.6% ± 39.8% decrease in consecutive push-ups able to be performed. Seventeen patients (38.6%) reported a degree of apprehension that affected their ability to lift weights. When all preoperative variables were accounted for, history of surgery to the contralateral shoulder (odds ratio, 0.600; 95% CI, 0.389-0.925) was associated with a decreased likelihood of returning to sport at the same or better level of intensity, while injury sustained during sport had a greater likelihood (odds ratio, 2.231; 95% CI, 1.234-4.031).

Conclusion: Patients undergoing PMT repair should expect significant functional improvements and a low complication rate. Yet, only 50% are able to return to preoperative intensity of sport, and they will also have significant reductions in their ability to weight lift.

Keywords: shoulder; weight lifting; muscle injuries

Rupture to the pectoralis major tendon (PMT) is an infrequent injury that is thought to occur during the lower phase of eccentric contraction as the PMT fibers are under tension.[21,23] Bench press is the classic motion associated with this injury, although this injury has been documented to occur during military maneuvers, football, and wrestling.[2,3,22] Furthermore, recreational resistance training is estimated to be practiced by 45 million adults in the United States, and reportedly, 1 in 5 adults aged 18 to 65 years participates in resistance training at least 2 times a week.[5] Given the increasing participation in weight lifting, clinicians may expect to encounter these soft tissue ruptures more frequently.

Operative management for ruptures is efficacious according to reported literature, particularly in acute settings, as ruptures to the myotendinous junction or tendon are generally amenable to anatomic repair.[2,8,9] Recent case series have demonstrated promising functional outcomes with regard to patient-reported outcome measures, return to sport, and ability to perform active military duties.[2,4,16,19] Yet, given the high percentage of athletes experiencing this injury, the ceiling effect may limit the utility of traditional outcome measures.

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The Bak criteria are the traditional measure of outcomes after pectoralis major repair (PMR).[2] This measure groups patients into 4 discrete categories based on percentage loss of isokinetic strength in comparison with preoperative status or the contralateral arm, cosmesis, pain, and return to sport without restriction. However, given the high functional demand of this group, sport involvement is highly variable; strength comparison with the contralateral arm may be biased by laterality; and performing routine and accurate testing in the clinical setting is not feasible. To better manage patient expectations, a thorough evaluation of pre- and postinjury metrics is needed to demonstrate the efficacy of PMR.

The purpose of the present study is to evaluate return to sport and return to weight lifting and to quantify percentage changes in common recreational weight-lifting exercises to better evaluate change in strength after PMT repair. The hypothesis is that patients who undergo PMR will experience significant reductions in strength secondary to the difficult nature of anatomically reducing the ruptured PMT.

METHODS

Patient Population

The design of the present study is a retrospective analysis of patients undergoing PMR for either acute (<6 months from injury) or chronic rupture of the PMT. A prospectively maintained institutional database was retrospectively queried for all patients who underwent PMR between January 2010 and August 2016 at a tertiary care center.

Data Collection

Following approval by the institutional review board, patients completed a standardized questionnaire to chart return-to-sport outcomes through an electronic data capture system (Outcomes Based Electronic Research Database; Universal Research Solutions, Columbia, MO, USA) after being contacted via telephone.[12-14,17]

These outcomes include preinjury sport participation, level of competition, postoperative sport participation, level of competition, and time to return to sport (see the Appendix, available in the online version of this article). American Shoulder and Elbow Surgeon (ASES) scores were collected preoperatively and at latest follow-up. Patient medical records were retrospectively reviewed for age at time of surgery, body mass index (BMI), workers' compensation status, and tear characteristics such as location, thickness, and head involvement. Additionally, the technique of surgical repair was also recorded.

Outcomes

Differences in preinjury and postoperative sport participation and level of competition were measured. Subanalysis of return to sports was performed with regard to contact sports, as classified by the American Academy of Pediatrics,[1] and overhead involvement. Additionally, patients were asked to self-report 1–repetition maximum (1RM) barbell bench press, 5–repetition maximum (5RM) barbell bench press, 1RM dumbbell bench press, 1RM dumbbell fly, and number of consecutive push-ups that they were able to perform within 3 years before their injury and at the latest follow-up date. The percentage change in these weight-lifting metrics was determined.

Surgical Technique

PMR was performed through an extended (4-6 cm) subpectoral incision with the patient in a modified beach-chair position. Both heads of the PMT were identified and followed to their tendinous insertion on the humerus. The insertion site of the torn PMT on the bicipital groove was debrided to facilitate implant placement. Between 1 and 4 implants of either double-loaded EndoButton (Pec Button; Arthrex) or double-loaded suture anchor (SutureFix; Smith & Nephew) were used for fixation. Implants were placed 2 cm apart from one another and loaded with No. 2 or No. 5 suture or suture tape (FiberWire, FiberTape; Arthrex or Cobraid; Smith & Nephew). The interval between the short head biceps and clavicular head of the PMT was identified and opened to visualize the rupture site. The stump of the PMT was stitched with either a running mattress or running Krackow stitch and subsequently tied down to create an anatomic reduction.

Statistical Analysis

RStudio (v 1.1.463; R Foundation for Computing) was utilized for statistical analysis. The Student *t* test was used to

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determine statistical differences in pre- and postinjury weight-lifting metrics and patient-reported outcomes. A stepwise multivariate logistic regression, with forward and backward selection, was performed to determine variables associated with returning to sport at the same or better intensity. Variables included in multivariate regression included age, BMI, history of surgery, dominant side affected, insurance status, tear location, partial versus complete thickness, head involvement, acute (<6 weeks from injury) versus chronic repair, surgical implant used, and injury mechanism. Significance was considered as $\alpha < .05$.

RESULTS

Patient Characteristics

Of 60 patients, 44 (73.3%) were available for follow-up. Mean ± SD follow-up was 51.1 ± 24.1 months (range, 24-103 months). All patients were male. Mean age at surgery was 39.6 ± 8.8 years, and mean BMI was 28.6 ± 3.5. The dominant side was affected in 20 of 44 cases. Acute repairs were performed in 30 cases, while 14 chronic cases were performed. The mean duration from injury to surgery was 3.8 ± 9.4 months. The mean duration from injury to survey was 55.4 ± 25.5 months. Five patients reported having previous surgery to the ipsilateral shoulder. Three reported a history of an ipsilateral PMR performed >6 months from the indexed procedure. One patient reported a rotator cuff repair, while another was unable to recall what surgery was performed.

Mechanism of Injury

Injury mechanisms were grouped into activities involving work, sport, weight lifting, or accidents. Sports wherein injury occurred included wrestling (twice), skiing, rock climbing, obstacle course running, water skiing, gymnastics, rugby, and general fitness at the gym. Work activities being performed wherein injury occurred included opening a trailer door, police officer duties (unspecified), pushing up oneself into an attic while firefighting, changing risers for maintenance, and opening a fire hydrant. Of those weight lifting, 21 of 26 patients were performing loaded bench press at the time of injury. One was performing a muscle-up, while the remaining 4 did not specify the exercise (Figure 1).

Tear Characteristics

Full-thickness ruptures were present in 31 of 44 patients, while the remainder were partial tears (Tietjen type II, 13; Tietjen type III, 31). The majority of patients had tears at the myotendinous junction (28 of 44; Tietjen type IIC, 9; Tietjen type IIIC, 19). Fourteen patients had tears at the tendon (Tietjen type IID, 4; Tietjen type IIID, 10). Two patients had tears to the muscle belly: one was also associated with a myotendinous junction tear (Tietjen type IIIC, 1), and the other had a tendon tear site (Tietjen type IIID, 1). The majority of patients had ruptures involving only the sternal head of the PMT (72.7%). There were 4.5% of

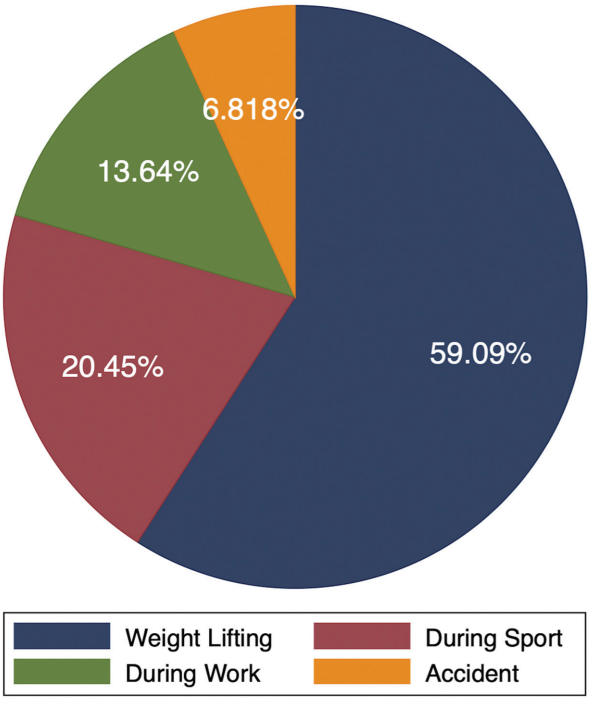

Figure 1. Activities performed during pectoralis major rupture.

patients who had injury exclusively to the clavicular head, while 22.7% had injury to the sternal and clavicular heads. Three patients had previously torn and repaired the PMT but experienced a rerupture of the tendon. Primary surgery in these patients was performed by an outside surgeon at more than a year from their second injury. Two patients also required utilization of allograft for anatomic reduction of their chronic tears. One of the patients who required allograft was also receiving revision surgery (Table 1).

Postoperative Outcomes

There were 3 complications within our cohort that required return to the operating room (6.8% complication rate). One patient had a rerupture of his PMT and underwent revision at an outside institution. Two patients had infection at the incision site. One of these patients underwent irrigation and debridement, while the other had concomitant implant removal in addition to the irrigation and debridement, as this infection occurred years later when the muscle was intact. All 3 patients were able to return to sport at final follow-up; however, 2 of 3 patients returned at a lower intensity. None of these patients reported apprehension in weight lifting. All 3 of these patients also reported good to excellent satisfaction with their surgery and physical status.

Overall, there were statistically significant improvements in both ASES and the Single Assessment Numeric Evaluation ($P < .001$, respectively). Good and excellent satisfaction was respectively reported in 22.7% and 56.8% of patients with regard to their surgical experience, 34.1%

TABLE 1
Patient and Tear Characteristics of Included Patients

	Mean ± SD or n (%)
Demographics	
Age, y	39.6 ± 8.8
Male:female	44:0
Body mass index	28.6 ± 3.5
Dominant-sided injury, yes:no	20:24
Previous surgery	
Contralateral (n = 6)	
Pectoralis major repair (outside institution)	4 (9.1)
Biceps tenodesis	1 (2.3)
Rotator cuff repair	1 (2.3)
Ipsilateral (n = 5)	
Pectoralis major repair (outside institution)	3 (6.8)
Rotator cuff repair	1 (2.3)
Unknown	1 (2.3)
Operative characteristics	
Thickness	
Partial	13 (29.6)
Complete	31 (70.4)
Location	
Myotendinous junction	28 (63.3)
Tendon	14 (34.1)
Muscle belly and tendon	1 (2.3)
Muscle belly and myotendinous junction	1 (2.3)
Head involvement	
Sternal	32 (72.7)
Clavicular	2 (4.5)
Both	10 (22.7)
Fixation	
Suture button	24 (54.6)
Suture anchors	15 (34.1)
Suture only	4 (9.1)
Bone tunnel and sutures	1 (2.3)
Augmented repair	
Allograft	2 (4.5)
Platelet-rich plasma	1 (2.3)

and 34.1% with regard to sporting activity, 40.9% and 25.0% with regard to overall physical fitness, and 25.0% and 40.9% with regard to cosmesis (Table 2).

Sporting Outcomes

All patients reported participation in some sport. Of 44 patients, 43 (97.7%) were able to return to sport at any level. Distribution of preinjury and final follow-up sports participation is demonstrated in Figure 2. In total, 22 patients (50.0%) reported returning to sport at the same or better intensity as preinjury status. Additionally, 28 (74.6%) reported same or better overall physical fitness at long-term follow-up in comparison with the preinjury level, while 16 (36.4%) reported worsened overall physical fitness. An independent subanalysis between partial/complete tears and the ability to return to sport at the same/better intensity was not significant (P = .741). Mean duration to full-duty sports was 7.7 ± 3.3 months after surgery. There were 3 outliers with respect to duration to full- duty sports. They returned to sport at 36, 42, and 50

TABLE 2
Postoperative Outcomes After Pectoralis Major Repair[a]

	Mean ± SD or n (%)
Complications	
Return to operating room	3 (6.8)
Infection: irrigation and debridement	1 (2.3)
Infection: irrigation and debridement + implant removal	1 (2.3)
Failure: revision pectoralis major repair	1 (2.3)
Sporting outcomes	
Apprehension with lifting	17 (38.6)
1RM barbell bench, lbs	
Preoperative	297.7 ± 81.0
Postoperative	212.2 ± 74.5
Percentage change, decrease	23.3 ± 45.6
5RM barbell bench, lbs	
Preoperative	246.3 ± 73.0
Postoperative	183.7 ± 62.6
Percentage change, decrease	14.7 ± 62.3
1RM dumbbell bench, lbs	
Preoperative	113.8 ± 51.8
Postoperative	79.8 ± 35.1
Percentage change, decrease	24.3 ± 21.8
1RM dumbbell fly, lbs	
Preoperative	55.2 ± 25.5
Postoperative	39.7 ± 18.9
Percentage change, decrease	35.7 ± 32.1
Consecutive push-ups, reps	
Preoperative	53.5 ± 27.5
Postoperative	41.0 ± 22.5
Percentage change, decrease	15.6 ± 39.8
Return to sport	43 of 44 (97.7)
Return to sport: same or better intensity	22 (50.0)
Change in overall fitness since surgery	
Better	6 (13.6)
Same	22 (50.0)
Worse	16 (36.4)
Good:excellent satisfaction	
Surgery	35 (79.5)
Excellent	25 (56.8)
Good	10 (22.7)
Sport	30 (68.2)
Excellent	15 (34.1)
Good	15 (34.1)
Fitness	29 (65.9)
Excellent	11 (25.0)
Good	18 (40.9)
Cosmesis	29 (65.9)
Excellent	18 (40.9)
Good	11 (25.0)
Patient-reported outcomes	
ASES	
Preoperative	63.4 ± 21.8
Postoperative	93.4 ± 10.2
SANE	
Preoperative	40.2 ± 23.6
Postoperative	90.9 ± 17.0

[a]ASES, American Shoulder and Elbow Surgeons; RM, repetition maximum; SANE, single assessment numeric evaluation.

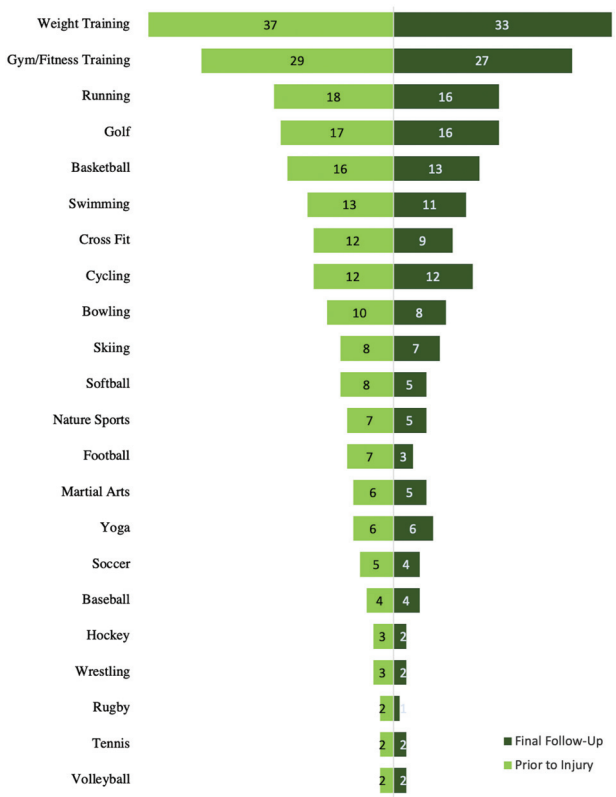

Figure 2. Sports participation before injury and at final follow-up after pectoralis major repair.

months postoperatively. The latter 2 patients reported apprehension when performing lifting activities. Before injury, there were 26 patients who reported participation in collision sports as the highest level of contact, while 19 patients reported collision sport participation at latest follow-up. Before injury, 6 patients reported that the highest level of contact was participation in limited-contact sports, while 8 reported doing so at latest follow-up. Before injury, 12 patients reported participating only in noncontact sports, while 17 reported doing so at latest follow-up. Additionally, 28 patients reported participation in sports with overhead involvement before injury, while 24 patients participated at final follow-up. There were statistically significant reductions in reported 1RM weight bench press ($P < .001$), 5RM bench press ($P < .001$), 1RM dumbbell bench press ($P = .001$), 1RM dumbbell fly ($P < .001$), and consecutive push-ups able to be performed ($P = .001$) (Table 3, Figure 3). Seventeen patients (38.6%) reported the degree of apprehension that affected their ability to lift weights. From subanalysis comparing partial and complete tears, there was no significant difference in proportion who were apprehensive from lifting (46.2% vs 35.5%, $P = .258$), reduction in 1RM bench press ($P = .631$), 5RM bench press ($P = .815$), 1RM dumbbell bench press ($P = .393$), 1RM dumbbell fly ($P = .829$), and consecutive push-ups ($P = .794$), respectively. There was no significant difference between partial and complete tears with regard to proportion of patients returning to sport at the same or better intensity (53.8% vs 43.4%, $P = .741$).

TABLE 3
Multivariate Stepwise Logistic Regression Analysis of Variables Associated With Returning to Sport at Same/Better Intensity

	Odds Ratio	95% CI	*P* Value[a]
Structure injured			
Clavicular head	1.536	1.144-4.221	.237
Sternal head	0.831	0.577-1.196	.327
Tendon	0.785	0.573-1.074	.140
Surgery side			
Contralateral	0.600	0.389-0.925	**.027**
Dominant	0.803	0.608-1.060	.131
Activity at injury			
Sport	2.231	1.234-4.031	**.012**
Work	1.538	0.799-2.962	.207
Lifting	1.112	0.799-2.962	.694
Acute repair[b]	0.745	0.608-1.060	.053

[a]Bold indicates $P < .05$.
[b]Performed within 6 weeks from injury.

Multivariate Analysis

When accounting for other variables, factors associated with increased likelihood of return to sport at the same or better intensity included sustaining an injury during sport ($P = .012$) and history of surgery to the contralateral shoulder ($P = .027$). Variables associated with decreased likelihood of returning to sport at the same or better level of function included history of surgery to the opposite shoulder (odds ratio [OR], 0.600; 95% CI, 0.389-0.925) and acute repairs (OR, 0.745; 95% CI, 0.559-0.993). Injury sustained during sport was associated with greater likelihood of returning to sport at the same or better level (OR, 2.231; 95% CI, 1.234-4.031). Apprehension was further evaluated for an independent relationship to acute tears; however, there was no significant relationship between these variables ($P = .786$).

DISCUSSION

The principal findings of the present study demonstrate a high return to sport (97.7%) after PMR at a mean follow-up of 51.1 months with a low incidence of returning to the operating room (6.8%). However, only 50% of patients were able to return to sport at the same or better level of function, with a high incidence of apprehension from lifting (38.6%). Significant reductions in all patient-reported weight lifting (barbell/dumbbell bench press, dumbbell fly, and consecutive push-ups) were observed within our cohort, which accounts for the ceiling effect otherwise present in legacy outcome measures.[2,15] The only variable associated with returning to sport at the same or better intensity was injuries sustained during sport. Patients with a history of surgery to the contralateral arm and acute repairs had a reduced likelihood of returning to sport at the same or better intensity. Despite the large overall success of this surgical procedure to return patients to activity, it is important that clinicians counsel patients regarding postoperative expectation with respect to changes in intensity when

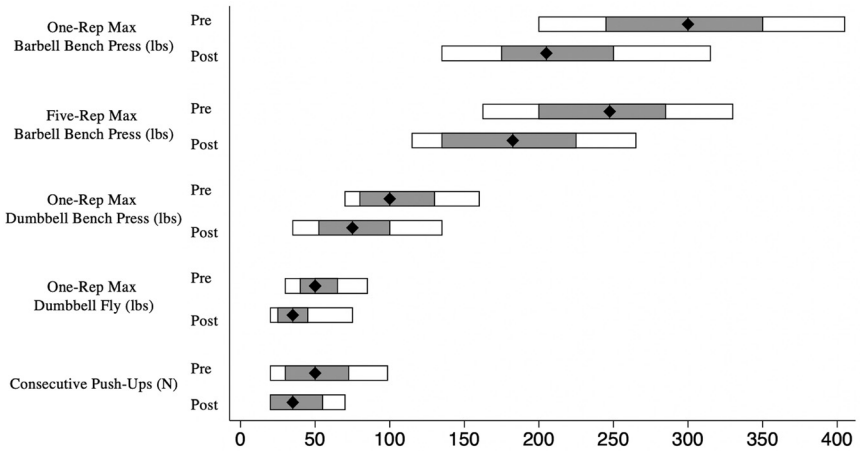

Figure 3. Distribution of weight-lifting outcomes within 3 years before injury (Pre) and at final follow-up (Post). Values are presented as median (diamond), interquartuartile range (gray), and range (white). Rep Max, repetition maximum.

returning to sport after PMR, especially given the high baseline function of this population.

Patients sustaining rupture of the PMT typically have high functional status because injury generally occurs during the eccentric contraction commonly seen in weight-lifting activities.[21,23] Mean preinjury bench press in this cohort and prior reports in literature is ≥300 lb, which attests to the high functional demand.[16] Previous reports relied on outcomes that may be limited by the ceiling effect, such as the ASES score, the Disabilities of the Arm, Shoulder and Hand score, the Short Form-36 score, and range of motion.[11,16,18,19,24] The Bak criteria are specific to PMT ruptures; however, functional improvement with respect to this score is based on symmetric and isokinetic strength change, which is challenging to measure in the clinical setting in this group.[2] To date, clinically significant outcome scores have not been established for PMR, which would serve as anchor points for predicted improvement after surgery.[6] Analysis of specific sports outcomes and weight-lifting outcomes offers a more comprehensive picture of expected postoperative outcome for this patient cohort, which has not been provided before in the literature.

Almost all patients within this cohort were able to return to sport and weight lifting (97.7%). However, a significant portion (38.6%) had apprehension with lifting, which may explain the decrease in maximum lifts across the board (barbell/dumbbell bench press, dumbbell fly, push-ups). As many of these injuries were sustained during weight lifting, there may be a psychological component holding these patients back despite advances in surgical technique and low retear rates (2.3%). However, this was not corroborated in multivariate analysis. Psychological comorbidity has been described as a negative predictor of outcome in orthopaedic literature, specifically in PMR.[19,20] Additionally, weight-lifting reductions are unsurprising after surgery, as repaired constructs have been shown to have reduced load to failure in comparison with the anatomic PMT.[10] Finally, the extended period of immobilization and recovery contributes to overall deconditioning that further complicates return to preinjury functional status in this age group. Continued evaluation of the influence of apprehension and mental status may lead to improved outcomes without increasing the risk of retear through controlled rehabilitation.

Results of the present study corroborate those of previous studies. Cordasco et al[7] reported a high rate of return to sports but did not report sport-specific return nor a breakdown of other weight-lifting activities besides the bench press. Additionally, the authors reported a decrease in bench press of nearly 40%, and although this was not a case-control series, this decrease was greater than that of the present series.[16] Mooers et al[18] described a limited series of 20 patients wherein 30% (n = 6) reported limited return to activity. However, it is unclear in what ways patients were limited in activity, and additional outcome metrics, such as outcome scores and Bak criteria, were reported in only 6 of the patients. Last, Nute et al[19] documented a 94% return to preoperative level of function within a mean 7.1 months after surgery in a cohort of military personnel. The present study demonstrated a lower return to same or better sport (50%) and a lengthier duration to this return (7.7 months), which may reflect differences between return to sport and return to military function. Assessing military duties is challenging, and there exists a degree of variability in the duties of military personnel. Furthermore, recreationally active patients may choose to give up a sport rather than return at a lower-level intensity, while this is less likely for military personnel. Additional external factors may also affect the return to military duties that we are unable to assess, such as rank, station, branch, and military occupational specialty not otherwise represented in sport. The remainder of the literature focuses on functional outcomes that are limited by the ceiling effect within this group.[2,11] The reported population is also older than that reported in recent studies, which may influence direct comparisons between groups.[1,16] Alternatively, review of the literature has demonstrated significantly reduced strength in nonsurgical management versus surgical groups (41.7% vs 14.3% reduction) and significantly

greater incidence of poor outcomes (31% vs 9.7%).[8] Nonoperative management, typically with sling and early mobilization, is therefore a viable option only for low-demand, elderly, or high–surgical risk cases, as the pectoralis major is not essential for activities of daily living.[4,21] Pooled data from a recent systematic review demonstrated 90% return to sport, 74% return to preinjury level of sport, and 95% return to work.[25] There is some discrepancy with respect to the present data; however, pooled analysis incorporates a great deal of heterogeneity from variable surgical technique, follow-up, and patient population.[25]

Limitations

The present study is not without its limitations. The study design was a retrospective review of a single-institution series of patients with a prospectively collected survey. Sporting outcomes, including weight-lifting metrics, were self-reported and may be subject to significant recall bias. Patients may also have had different modes of rehabilitation (eg, professional physical therapy vs self-modulated), which may bias outcomes of the surgery. Additionally, the population cohort is heterogeneous with respect to injury pattern and surgical technique, although this may be more applicable to a larger population. Subanalysis by partial versus complete pectoralis tears and multivariate regression analysis were performed to account for this heterogeneity; however, relationships may still be present that can be observed only in larger populations.

CONCLUSION

Patients undergoing PMT repair should expect significant functional improvements and a low complication rate. Yet, only 50% are able to return to preoperative intensity of sport, and they will also have significant reductions in their ability to weight lift.

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